SHUBHAM PRAJAPATI

Backend Engineer • Distributed Systems • gRPC & Event-Driven Architecture • Real-Time Systems

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PROFESSIONAL SUMMARY

Results-driven Backend Engineer with 3+ years of production experience building high-performance, distributed systems in Golang. Deep expertise in designing gRPC and REST APIs, event-driven architectures, and real-time WebSocket systems that scale under concurrent load. Skilled at architecting microservices with clean separation of concerns, async pub-sub pipelines, and robust authentication layers (JWT, PASETO, OAuth 2.0). Comfortable owning the full backend lifecycle — from database schema and API design through containerized deployment, CI/CD automation, and production infrastructure management on AWS, GCP, and Vultr. Open source contributor with hands-on experience in decentralized and peer-to-peer systems (libp2p, DePIN networks).

TECHNICAL SKILLS

Languages	Golang (primary), SQL, Bash
APIs & Protocols	REST, gRPC (Protocol Buffers), WebSocket, HTTP/2
Distributed Systems	Microservices, libp2p, DePIN networks, pub-sub pipelines, async event architectures
Event-Driven Arch.	Pub-Sub messaging, Redis Streams, async pipelines, producer-consumer decoupling
Real-Time Systems	WebSocket servers, live data streaming, concurrent connection management
Auth & Security	JWT, PASETO v4, OAuth 2.0, OIDC, Ethereum wallet challenge-response, SSL/TLS
Databases	PostgreSQL, MySQL, gORM, ElephantSQL, Supabase, Redis, TimescaleDB, Valkey
Containers & DevOps	Docker, Docker Compose, GitHub Actions CI/CD, container networking
Cloud & Infra	AWS, GCP, Vultr — provisioning, server management, Traefik, Cloudflare (WAF, Workers, Zero Trust), Uptime Kuma
Networking	WireGuard, iptables NAT, DNS (PiHole), wgctrl, TCP/IP, SSH
Tools	Postman, Git, Jira, Notion, Slack, Discord

PROFESSIONAL EXPERIENCE

Backend Engineer | Intellixente Solutions — Pune, India (On-site) *Apr 2023 – Present*

Project: X-Factor — Enterprise Project Management System | Golang, PostgreSQL, Mux, Docker

- Architected scalable, multi-tenant backend infrastructure in Golang — owning the full stack from database schema design through API layer and containerized deployment.
- Engineered REST APIs for user authentication (JWT/PASETO), role-based access control, and Excel-based project plan ingestion with robust server-side validation.
- Designed and implemented an event-driven architecture for real-time task status propagation — async pub-sub pipelines decoupling producers from consumers, ensuring reliable delivery under concurrent load.
- Implemented superadmin dynamic company context switching — a stateless JWT design embedding company scopes, enabling seamless multi-tenant isolation without session state.
- Managed Docker networking, container inspection, and server infrastructure on Vultr; configured SSL/TLS hardening (HSTS, redirect middleware) behind Cloudflare and Traefik.

- Led Git branching strategy, PR reviews, and GitHub Actions CI/CD pipelines for automated testing and zero-downtime deployment.

Backend Engineer & Server Handler | Lazarus Network — Florida, USA (Remote) *Mar 2022 – Mar 2023*

High-performance event/community platform with Blockchain, NFT minting, and virtual event capabilities

- Led full project lifecycle from backend infrastructure setup through production deployment, including OAuth-authenticated REST API and gRPC service design for internal microservice communication.
- Designed and implemented an event-driven architecture powering ticket purchasing, virtual event streaming, and NFT minting — using async pub-sub pipelines to handle bursty, high-concurrency workloads gracefully.
- Implemented Redis caching layer for hot-path API endpoints, achieving a 50% reduction in API response times under high-traffic event loads.
- Built WebSocket-based real-time notification system for live event updates and community activity feeds, supporting thousands of concurrent subscribers.
- Integrated Blockchain/NFT minting pipelines, Stripe payments, and MailGun transactional email — orchestrating multiple distributed services reliably.
- Managed server infrastructure, ensuring high availability, monitoring, and rapid bug fixes under production load.
- Tech stack: Golang, gORM, PostgreSQL, Gin Gonic, gRPC, WebSocket, Redis, Docker, Supabase, Stripe, MailGun, GitHub Actions.

Sotreus — Decentralized VPN Infrastructure | Lazarus Network

WireGuard orchestration, DNS firewall, kernel networking — Golang, Docker, wgctrl, PiHole

- Built a multi-instance WireGuard VPN orchestration system supporting isolated VPN + DNS firewall pairs per region, enabling secure and ad-filtered tunneling for all connected clients.
- Designed and implemented gRPC and REST APIs in Golang for programmatic VPN client provisioning, integrated with PASETO v4 token authentication and Ethereum wallet challenge-response.
- Engineered iptables NAT rules, DNS interception via PiHole, and kernel-level WireGuard interface management using wgctrl.
- Automated multi-instance deployment using Docker Compose with dynamic port allocation, duplicate instance prevention, and per-instance volume isolation.

OPEN SOURCE PROJECTS

Netsepio — Decentralized Internet Infrastructure (DePIN)

Golang, PostgreSQL, Docker, libp2p, gRPC, Next.js, BullMQ, Valkey, Cloudflare WAF & Workers

- Contributed to a decentralized physical infrastructure network (DePIN) designed for secure, unrestricted global connectivity.
- Worked with libp2p for peer-to-peer networking and gRPC for inter-node service communication, managing distributed node routing and protocol design in Golang and Docker.
- Engaged with real-world challenges of building overlay networks, secure tunnels, and high-availability distributed endpoints.
- Secured and operated production infrastructure for a NetSepio ecosystem platform (Next.js web app, mobile API, BullMQ workers) running on Vultr behind Cloudflare.
- Investigated abnormal traffic of 1.01M requests per day, 72% from a single scraper IP, and deployed per-IP API rate limiting within Cloudflare Free-plan limits, validated by load testing.
- Built a Cloudflare Worker on a cron trigger that queries the GraphQL Firewall Events API and adds repeat rate-limit offenders to a WAF-enforced IP block list.
- Found and closed a WAF bypass where spoofable Origin-header skip rules let any client evade bot protection; replaced them with IP-based and secret-header rules for monitoring and mobile clients.
- Diagnosed a production outage caused by an ioredis/BullMQ connection leak that had grown to 7,817 of 7,848 maximum Valkey clients over 31 days.
- Designed PostgreSQL access control across production, development and test databases: role isolation, scoped grants without DELETE/TRUNCATE for human users, default privileges and Row Level Security policies.

- Set up uptime, memory and disk alerting with Uptime Kuma push monitors and Teams notifications, using Cloudflare Access service tokens so Zero Trust no longer silently blocked heartbeats.

MyriadFlow — NFT / Phygital Platform

Golang, PostgreSQL, Docker, gRPC

- Built and maintained backend services for an NFT marketplace handling high-throughput event-driven workflows — async event pipelines for mint triggers, ownership transfers, and marketplace activity.
- Contributed to gRPC-based inter-service communication, microservices architecture design, and database optimization for scalable data retrieval under concurrent load.

PERSONAL PROJECTS

Atlas — Fleet Observability Platform

Golang, libp2p, PostgreSQL + TimescaleDB, React/TypeScript, Docker, GitHub Actions — personal project, deployed to monitor a live server fleet

- Built a fleet observability platform end to end: Go agents on remote servers stream host metrics to a central control plane over libp2p, stored in PostgreSQL + TimescaleDB and served through a React/TypeScript dashboard.
- Replaced bearer enrollment tokens with libp2p Peer ID–based mutual authentication and a per-environment allowlist, with a unique persistent keypair per agent and per relationship (local vs. production).
- Added a protected superadmin tier to the role-based access model (viewer, operator, admin): admins cannot modify the superadmin across all 7 user-management actions, verified in production with no regression for admin-to-admin actions.
- Built a GitHub Actions release pipeline with a hard gate that blocks dirty builds from becoming release artifacts, and tag rules that keep test tags out of production and off the :stable image.
- Wrote idempotent agent install and migration scripts with dry-run mode, a 12-point health check, byte-identical identity verification and automatic rollback on failure.
- Recovered the metrics database after a full disk left TimescaleDB compression jobs stuck: rebuilt compression policies and chunk intervals, cutting the database from 17 GB to 1.1 GB.

Real-Time Trading Calculator — WebSocket Engine

Golang, WebSocket — Live buy/sell signal streaming with P&L computation

- Developed a real-time WebSocket server for a live trading buy/sell calculator — streaming market signals, computing position P&L, and broadcasting live updates to multiple concurrent clients with sub-millisecond latency.
- Designed stateful connection management to track per-client session context, handle reconnections gracefully, and maintain consistent order book state across concurrent subscribers.

KEY STRENGTHS

- Distributed systems architecture: Hands-on experience designing microservices, pub-sub event pipelines, gRPC inter-service communication, and peer-to-peer networks (libp2p, DePIN) — building systems that stay reliable under concurrent, high-throughput load.
- Event-driven design: Consistently chosen async event architectures across production projects to decouple services, improve resilience, and handle bursty workloads without bottlenecks.
- API depth: Fluent in both REST and gRPC — understands the trade-offs and picks the right protocol per use case; experienced with Protocol Buffers, HTTP/2, and WebSocket for different latency and throughput requirements.
- Performance engineering: Proven track record of measurable improvements — 50% API response time reduction with Redis caching; sub-millisecond WebSocket broadcast latency in the trading calculator.
- Security-first mindset: JWT, PASETO v4, OAuth 2.0, Ethereum wallet auth, and SSL/TLS hardening applied at every layer — not bolted on after the fact.
- Full ownership mentality: Comfortable taking a feature from schema design → API implementation → containerized deployment → CI/CD pipeline → production monitoring.
- Self-driven learner: Transitioned from a Mechanical Engineering background into backend engineering through open-source contributions and production project work — fast to pick up new domains and technologies.

EDUCATION

Bachelor of Engineering — Mechanical Engineering *Graduated 2022*
Gandhi Institute for Technology (GIFT), Bhubaneswar, Odisha